

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Differential Equations (DE) :

An equation involving derivatives of one or more dependent variable with respect to one or more independent var is called a differential equation.

Ordinary Differential Equation (ODE) :

A differential equation involving ordinary derivatives of one or more dependent variables with respect to a single independent variable is called an ordinary differential equation.

Partial Differential Equation (PDE) :

A differential equation involving partial derivatives of one or more dependent variable with respect to more than one independent variable is called a diff partial Differential Equation.

Examples: $\frac{d^2 y}{dx^2} + xy \left(\frac{dy}{dx} \right)^2 = 0 \rightarrow \text{ODE}$

$$\frac{d^4 x}{dy^4} + 5 \frac{d^2 x}{dy^2} + 3x = \sin y$$

$$\frac{\partial v}{\partial s} + \frac{\partial v}{\partial t} = v, \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

PDE

$$\frac{d}{dx}(e^{ax}) = ae^{ax}, \quad \frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

Not DE but Derived.

Order of a DE:

The order of the highest ordered derivative involved in a differential equation is called the order of the differential equation.

Degree of DE: The power (or degree) of the highest ordered derivatives involved in a differential equation is called the degree of the differential equation.

Example: $\frac{d^2 y}{dx^2} + xy \left(\frac{dy}{dx}\right)^2 = 0 \rightarrow$ 2nd order 1st degree ODE

$\frac{d^4 x}{dt^4} + 5 \frac{d^2 x}{dt^2} + 3x = \sin(t) \rightarrow$ 4th order 1st degree ODE

$\frac{\partial v}{\partial s} + \frac{\partial v}{\partial t} = v \rightarrow$ 1st order 1st degree PDE

$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 \rightarrow$ 2nd order 1st degree PDE

$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 \rightarrow$ 2nd order 1st degree PDE

Linear ODE: A linear ordinary differential equation of order n , in the dependent variable y and the independent variable x , is an equation that can be expressed in the form,

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \frac{dy}{dx} + a_n(x) y = b(x)$$

where a_0 is not identically zero

Observations

- 1) The dependent variable y and its various derivatives occur to the 1st degree only.
- 2) No products of y and/or any of its derivatives are present.
- 3) No transcendental functions of y or its derivatives occur.

Non-Linear ODE: A non linear ordinary differential equation is an ordinary differential

equation, that is not linear.

Example: $\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = 0$ — (a)

linear ODE $\left[\frac{d^4 y}{dx^4} + x^2 \frac{d^3 y}{dx^3} + x^3 \frac{dy}{dx} = x e^x \right]$ — (b)

eqn (a) is linear with constant (1, 5, 6) coefficients
eqn (b) is " " " " variable (1, x^2 , x^3) " "

$$\# \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y^2 = 0$$

The dependent variable y is raised to the second order.

$$\frac{d^2 y}{dx^2} + 5 \left(\frac{dy}{dx} \right)^3 + 6y = 0$$

Non linear ODE

$$\frac{d^2 y}{dx^2} + 5y \frac{dy}{dx} + 6y = 0$$

The dependent variable y and its derivative is multiplied together.

Origin and application: DEs occur in connection with

numerous problems that are encountered in the various branches of science and engineering.

e.g. Population Dynamics, Dynamical System,

Fluid Dynamics, Meteorology, Heat Transfer, Bio-Physics, etc.

Formation of DE: Form the DE from $y = ax + bx^2$
Eliminate the constants.

Sol: we have, $\frac{dy}{dx} = a + 2bx$; $\frac{d^2 y}{dx^2} = 2b$ (1)

$$\Rightarrow a = \frac{dy}{dx} - 2bx = \frac{dy}{dx} - 2x \frac{d^2 y}{dx^2}$$

$$\therefore (1) \Rightarrow y = x \left(\frac{dy}{dx} - 2x \frac{d^2 y}{dx^2} \right) + \frac{1}{2} x^2 \frac{d^2 y}{dx^2}$$

$$\Rightarrow y = x \frac{dy}{dx} - x^2 \frac{d^2 y}{dx^2} + \frac{1}{2} x^2 \frac{d^2 y}{dx^2}$$

$$\Rightarrow x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0 \text{ which is a DE of 2nd order.}$$

7) General Solution: The relation containing n -arbitrary constants which satisfies an ODE of n th order is called it's called complete primitive or general solution.

Note: It can be shown that, by eliminating n arbitrary constants from an eqn in x, y , we get a DE of n th order. Such a process is called transformation of DE.

*) Particular Solution: A particular solⁿ is one obtained from the primitive or general solⁿ by assigning definite values to the arbitrary constants.

Geometrically, the primitive is the eqn of family of curves satisfying the DE and a particular solⁿ is the eqn of the some one of this family of curves.

Ex: Form the DE from family of parabolas

(1) $(x-h)^2 = 4a(y-k)$ — (1), a, h, k constants

Diff (1) $\Rightarrow 2(x-h) = 4a \frac{dy}{dx} \mid \frac{dy}{dx} = 0$

Diff again, $\Rightarrow 2 = 4a \frac{d^2y}{dx^2} \Rightarrow \frac{d^2y}{dx^2} = \frac{1}{2a}$

$\frac{d^3y}{dx^3} = 0$ which is the required DE.

Nature of Solution: Consider the n th order DE:

$$F\left[x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}\right] = 0 \quad \text{--- (1)}$$

where F is a real function of $(n+2)$ arguments

$$x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}$$

[1] Let f be a real f^n defined for all x in an interval I and having an n th derivative for all $x \in I$. The $f^n f$ is called an explicit solⁿ of the DE (1) on I if it fulfills the following 2 requirements,

(i) $F[x, f(x), f'(x), \dots, f^{(n)}(x)]$ is defined

for all $x \in I$.

(ii) $F[x, f(x), f'(x), \dots, f^{(n)}(x)] = 0 \quad \forall x \in I$

[2] A relation $g(x, y) = 0$ is called an implicit solution of the DE (1) if this relation defines at least one real function f of variable x on I such that this function is an explicit solution of (1) on I .

③ Both explicit and implicit solutions will usually be called simply solutions.

Ex: Show that, $f(x) = 2 \sin x + 3 \cos x$ is an explicit solⁿ of $\frac{d^2 y}{dx^2} + y = 0$ for all/every value of x

Solⁿ: $f(x) = 2 \sin(x) + 3 \cos(x)$, $f''(x) = -2 \sin(x) - 3 \cos(x)$

$$f'(x) = 2 \cos(x) - 3 \sin(x)$$

$$\text{Now, } \frac{d^2 y}{dx^2} + y = -2 \sin(x) - 3 \cos(x) + 2 \sin(x) + 3 \cos(x) = 0$$

which holds for all real x . The f is an explicit

solⁿ of the DE $y'' + y = 0$

Ex: Consider $\frac{dy}{dx} = 2x$ - ①
DE ① has a one-parameter family of solⁿ of the form $y = x^2 + c$ - ②, c is arbitrary constant or parameter of the family



Ex: Show that, $f(x) = c_1 e^{4x} + c_2 e^{-2x}$, c_1, c_2 are arbitrary
 $f'(x) = 4c_1 e^{4x} - 2c_2 e^{-2x}$
 $f''(x) = 16c_1 e^{4x} + 4c_2 e^{-2x}$

DE ① we get, $y'' - 2y' - 8y$

$$= 16c_1 e^{4x} + 4c_2 e^{-2x} - 2(4c_1 e^{4x} - 2c_2 e^{-2x}) - 8(c_1 e^{4x} + c_2 e^{-2x})$$

$$= 16c_1 e^{4x} + 4c_2 e^{-2x} - 8c_1 e^{4x} + 4c_2 e^{-2x} - 8c_1 e^{4x} - 8c_2 e^{-2x} = 0$$

$\Rightarrow$ f is an explicit of the DE ① for all x

Initial Value Problem, Boundary Value Problem,

Existence of Solution

Example: $\frac{dy}{dx} = 2x \dots$ ①, $y(1) = 4 \dots$ ②

Eqn ① has a one-parameter family of solutions

such as, $y = x^2 + c \dots$ ③, c is an arbitrary const.

By ②, put $x = 1$, then from ③ $\Rightarrow$

$$y(1) = 1^2 + c \Rightarrow 4 = 1 + c \Rightarrow c = 3$$

③ $\Rightarrow y = x^2 + 3$ which is indeed a solⁿ of ① satisfying the condition ②

$\Rightarrow$ In the application of both first and higher order D.E., they involve with a D.E and one or more supplementary conditions which the solution of the given D.E must satisfy.

$\Rightarrow$ If all of the associated supplementary conditions relate to one x value, the problem is called an initial-value problem.

(or One point-boundary value Problem)

⇒ If the conditions relate to two different x values
 BVP the problem is called two point boundary value problem.
 simply boundary value problem.

Ex: $\frac{d^2y}{dx^2} + y = 0, [y(1) = 3, y'(1) = 4]$ I.V.P.

Initial Conditions

Ex: $\frac{d^2y}{dx^2} + y = 0, [y(0) = 1, y(\frac{\pi}{2}) = 5]$ BVP

Boundary Conditions

Existence and Uniqueness Theorem: Consider the DE

where, $f(x, y)$ is continuous f'' in some domain D of the xy plane and, $\frac{dy}{dx} = f(x, y)$ (*)

(2) $\frac{\partial f}{\partial y}$ is also contⁿ f'' of x & y and let, (x_0, y_0) be a point D

let, (x_0, y_0) be a point D

⇒ There exists a unique solⁿ ϕ of the DE (*) defined on some interval $|x - x_0| \leq h$, where h is sufficiently small, that satisfy the condition,

$\phi(x_0) = y_0$

P.S.: Please read Page # 15 - P# 21 (J.L. Ross)

Formation of DE's

Eliminating ordinary arbitrary constants:

Ex: Form the DE by eliminating arbitrary constants from $y = ax + bx^2$.

Solⁿ: Given, $y = ax + bx^2$ --- (1)

Diff, $\frac{dy}{dx} = a + 2bx$ --- (2)

Again diff $\Rightarrow \frac{d^2y}{dx^2} = 2b$ --- (3)

--- (4) $\frac{d^2y}{dx^2} = 2b$

(2) $\Rightarrow a = \frac{dy}{dx} - 2bx = \frac{dy}{dx} - \frac{d^2y}{dx^2} x$

by (3) and (4), (1) $\Rightarrow y = \left(\frac{dy}{dx} - \frac{d^2y}{dx^2} x \right) x + \frac{1}{2} \frac{d^2y}{dx^2} x^2$

$\Rightarrow x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$ which is the required

DE of 2nd order

Form the DE of all parabolas whose axes are parallel to the axis of y.

Solⁿ: The eqn of the given parabolas can be written as: $(x-h)^2 = 4a(y-k)$ --- (1)

where a, h, k are arbitrary constants.

Diff $\Rightarrow 2(x-h) = 4a \frac{dy}{dx} \Rightarrow (x-h) = 2a \frac{dy}{dx}$

It again $\Rightarrow 1 = 2a \frac{d^2y}{dx^2}$ i.e. $\frac{d^2y}{dx^2} = \frac{1}{2a}$

10 once again, $\frac{d^3y}{dx^3} = 0$ which is the required DE.

Equation of first order and 1st degree.

The 1st order DE: $\frac{dy}{dx} = f(x, y)$ (i) (derivative form)

or, $M(x, y) dx + N(x, y) dy = 0$ (ii) (differential form)

An eqn. in one of these forms may readily be written in the other form.

Separable Eqns and eqns reducible to this form:

* An eqn of the form, $F(x) G(y) dx + f(x) g(y) dy = 0$

is called an equation with variables separable or

simply a separable equation.

Ex: Solve $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$

Soln: Separating the variables, we get,

$$\frac{\sec^2 x}{\tan x} dx + \frac{\sec^2 y}{\tan y} dy = 0 \quad \text{Integrating} \Rightarrow \log \tan x + \log \tan y = C$$

$$\Rightarrow \log (\tan x \tan y) = C \Rightarrow \tan x \tan y = e^C = C_1$$

Solve the IVP: $x \sin y dx + (x^2 + 1) \cos y dy = 0$ (I)

with IC $y(1) = \frac{\pi}{2}$ (II)

Soln: Separating the variables $\Rightarrow$

$$\frac{x}{x^2 + 1} dx + \frac{\sin y}{\cos y} dy = 0 \quad [\text{assuming } y \neq 0]$$

$$\text{Integrating} \Rightarrow \frac{1}{2} \ln(x^2 + 1) - \ln(\cos y) = \ln C$$

$$\Rightarrow \ln(x^2+1) + 2\ln(\sin y) = 2\ln c \Rightarrow \ln[(x^2+1)(\sin^2 y)] = \ln c$$

$$(x^2+1)(\sin^2 y) = c \quad \text{--- (III)}$$

By (II) and (III) $(x^2+1) \sin^2 y = c \Rightarrow c \geq 2$

(III) $\Rightarrow (x^2+1) \sin^2 y = 2$ which is the required solⁿ.

Eqn^s reducible to the form of variable separable.

Ex: Solve $\frac{dy}{dx} = (4x+y+1)^2$ --- (1)

Solⁿ: Putting, $4x+y+1 = u \Rightarrow 4 + \frac{du}{dx} = \frac{du}{dx}$

The eqn (1) reduces to, $\frac{du}{dx} - 4 = u^2$

$$\Rightarrow \frac{du}{dx} = u^2 + 4 \Rightarrow \frac{du}{u^2 + 2^2} = dx \quad \xrightarrow{\text{Integ}} \quad \frac{1}{2} \tan^{-1} \frac{u}{2}$$

[which is in the form of variable separable] $= x + C$

$$\Rightarrow \frac{1}{2} \tan^{-1} \left(\frac{4x+y+1}{2} \right) = x + C$$

Ex: $(x+y)^2 \frac{dy}{dx} = a^2$ --- (1)

Solⁿ: Put $x+y = u \Rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{du}{dx} - 1$

$$(1) \Rightarrow u^2 \frac{dy}{dx} = a^2 \Rightarrow u^2 \left(\frac{du}{dx} - 1 \right) = a^2$$

$$\Rightarrow \frac{du}{dx} = \frac{a^2}{u^2} + 1 = \frac{a^2 + u^2}{u^2} \quad \therefore \frac{u^2}{a^2 + u^2} du = dx$$

$$\Rightarrow \left(1 - \frac{a^2}{a^2 + u^2} \right) du = dx$$

Integrating $\Rightarrow u - \frac{a^2}{a} \tan^{-1} \frac{u}{a} = x + C$

$$\Rightarrow (x+y) - a \tan^{-1} \frac{x+y}{a} = x+c \quad \therefore y = a \tan^{-1} \frac{x+y}{a} + c$$

Ans:

Homogenous DE: The 1st order DE $M(x,y)dx + N(x,y)dy = 0$ is said to be homogenous if it's written in the form $\frac{dy}{dx} = \frac{f_1(x,y)}{f_2(x,y)}$ in which $f_1(x,y)$ & $f_2(x,y)$ are homogenous fⁿ of the same degree.

The homogenous eqn can be reduced to an eqn in which variables are separable by putting

$$y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

[If $\frac{dy}{dx} = f(x,y)$ then it is homogenous if $f(x,y)$ can be written in the form $g\left(\frac{y}{x}\right)$]

Ex: Solve $(x^2 + y^2)dx + 2xydy = 0$ — (1)

Solⁿ: (1) $\Rightarrow \frac{dy}{dx} = -\frac{x^2 + y^2}{2xy}$ (homogenous)

$\therefore f(x,y) = -\frac{x^2 + y^2}{2xy}$, $f(kx, ky) = -\frac{(kx)^2 + (ky)^2}{2kxky}$

$\Rightarrow n=0$ (zero degree) $= -\left(\frac{x^2 + y^2}{2xy}\right)$

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$ — (2)

by (2) (1) $\Rightarrow v + x \frac{dv}{dx} = \frac{x^2 + x^2 v^2}{2x \cdot vx}$

$\Rightarrow v + x \frac{dv}{dx} = \frac{1 + v^2}{2v} \Rightarrow x \frac{dv}{dx} = \frac{1 + v^2}{2v} - v$

$\Rightarrow \frac{dv}{\frac{1 + v^2}{2v} - v} = \frac{dx}{x}$ (variable separable)

Integrating $\Rightarrow \ln x = \frac{-1}{3} \ln(1 + 3v^2) + \ln C$

$\Rightarrow \ln x + \ln(1 + 3v^2)^{1/3} = \ln C$
 $\Rightarrow x(1 + 3v^2)^{1/3} = C$ Ans.

Ex: $(x^2 - 3y^2) \frac{dx}{dy} + 2xy = 0$

$\Rightarrow \frac{dx}{dy} = \frac{-2xy}{x^2 - 3y^2} = -\frac{x}{2y} + \frac{3}{2} \frac{y}{x}$

$= \frac{3}{2} \left(\frac{y}{x}\right)$ (homogeneous, since $\frac{y}{x}$)

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$

we get, $v + x \frac{dv}{dx} = \frac{-1}{2v} + \frac{3}{2} v$

$\Rightarrow x \frac{dv}{dx} = \frac{3}{2} v - v - \frac{1}{2v} = \frac{1}{2v} + \frac{1}{2} v$

Integrating $\ln(x) = \ln|v^2 - 1| + \ln C - \ln C$

$\Rightarrow v^2 - 1 = Cx \Rightarrow \left(\frac{y}{x}\right)^2 - 1 = Cx$

$\therefore y^2 - x^2 = Cx^3$ Ans.

Ex: $y + \sqrt{x^2 + y^2} dx - x dy = 0, y(1) = 0$ S.L. Russ, P. 46

Solⁿ: Given, $y + \sqrt{x^2 + y^2} dx - x dy = 0$

$$\Rightarrow y + \sqrt{x^2 + y^2} dx = x dy - y$$

$$\Rightarrow \sqrt{x^2 + y^2} dx = x \frac{dy}{dx} - y$$

$$\Rightarrow \frac{dy}{dx} = \left(\sqrt{x^2 + y^2} + \frac{y}{x} \right) \frac{1}{x}$$

Ans: Let $y = vx$, then $x = \frac{y}{v}$, $dx = \frac{y dv - v dy}{v^2}$

$$\frac{dy}{dx} = \frac{v + x \frac{dv}{dx}}{1} = \frac{v + x \frac{dv}{dx}}{1}$$

$$\frac{v + x \frac{dv}{dx}}{1} = \left(\sqrt{x^2 + y^2} + \frac{y}{x} \right) \frac{1}{x}$$

Substituting $y = vx$ and $dx = \frac{y dv - v dy}{v^2}$ into the equation:

$$\frac{v + x \frac{dv}{dx}}{1} = \left(\sqrt{x^2 + y^2} + \frac{y}{x} \right) \frac{1}{x}$$

$$\frac{v + x \frac{dv}{dx}}{1} = \left(\sqrt{x^2 + y^2} + \frac{y}{x} \right) \frac{1}{x}$$

$$\frac{v + x \frac{dv}{dx}}{1} = \left(\sqrt{x^2 + y^2} + \frac{y}{x} \right) \frac{1}{x}$$

Reducible to homogenous form!

Eqn of the form $\frac{dy}{dx} = \frac{ax+by+c}{a_1x+b_1y+c_1}$, $\frac{a}{a_1} \neq \frac{b}{b_1}$

can be reduced to homogenous form by putting

$$x = X+h, y = Y+k \text{ then, } \frac{dy}{dx} = \frac{dY}{dX}$$

Ex: Solve, $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$ --- ①

Solⁿ: Put, $x = X+h, y = Y+k$, h, k constant

$$\text{Then, } \frac{dy}{dx} = \frac{dY}{dX} \therefore \text{①} \Rightarrow \frac{dY}{dX} = \frac{X+h+2(Y+k)-3}{2(X+h)+(Y+k)-3}$$

Now, choose h, k so that,

$$h+2k-3=0 \text{ and,}$$

$$2h+k-3=0, \text{ Solving these}$$

we get, $h=1, k=1$

$$\therefore \frac{dY}{dX} = \frac{X+2Y}{2X+Y} \text{ which is homogenous in } X \text{ and } Y$$

$$\text{Put, } Y = vX \text{ so that, } \frac{dY}{dX} = v + X \frac{dv}{dX}$$

$$\Rightarrow v + X \frac{dv}{dX} = \frac{X+2vX}{2X+vX} = \frac{1+2v}{2+v}$$

$$\Rightarrow X \frac{dv}{dX} = \frac{1+2v}{2+v} - v = \frac{1-v^2}{2+v}$$

$$\Rightarrow \frac{dX}{X} = \left(\frac{2+v}{1-v^2} \right) dv = \left(\frac{2}{1-v^2} + \frac{v}{1-v^2} \right) dv$$

$$\Rightarrow \ln X = 2 \cdot \frac{1}{2} \ln \frac{1+v}{1-v} - \frac{1}{2} \ln |1-v^2| + \ln c$$

$$\Rightarrow X = c \left(\frac{1+v}{1-v} \cdot \frac{1}{\sqrt{1-v^2}} \right) = c \cdot \frac{1+v}{1-v} \cdot \frac{1}{\sqrt{1+v} \sqrt{1-v}}$$

$$\Rightarrow X = \frac{c \sqrt{1+v}}{(1-v)^{3/2}} \Rightarrow X^2 (1-v)^3 = c^2 (1+v)$$

$$\Rightarrow X^2 \left(1 - \frac{y}{x} \right)^3 = c^2 \left(1 + \frac{y}{x} \right) \quad \left[\because v = \frac{y}{x} \right] \Rightarrow (x-y)^3 = c^2 (x+y)$$

$$\Rightarrow \{(x-1) - (y-1)\}^3 = c^2 \{(x-1) + (y-1)\} \quad \left[\because x = x+1, y = y+1 \right]$$

$$\Rightarrow (x-y)^3 = c^2 (x+y-2) \text{ which is the required soln.}$$

Solved Problems and Exercises of SL Rom / BD Sharma

Linear DE: A DE of the form $\frac{dy}{dx} + P(x)y = Q(x)$

is called the linear DE of the 1st order.

To solve (1), multiply both sides by $e^{\int P(x) dx}$

Then it becomes, $e^{\int P(x) dx} \frac{dy}{dx} + P(x)y e^{\int P(x) dx} = Q(x) e^{\int P(x) dx}$

$$\Rightarrow \frac{d}{dx} (y e^{\int P(x) dx}) = Q(x) e^{\int P(x) dx}$$

Integrating with respect to $x \Rightarrow$

$$y e^{\int P(x) dx} = \int Q(x) e^{\int P(x) dx} + c$$

which is the required soln.

Integrating factor: It's note that to solve (1),

we multiply it by a factor $e^{\int P(x) dx}$ and

equation becomes readily / directly integrable.

Such a factor is called integrating factor (IF)

Ex: Solve, $x \frac{dy}{dx} + 2y = x^2 \log x$ --- ①

Sol: ① $\Rightarrow \frac{dy}{dx} + \frac{2}{x}y = x \log x$

$\therefore IF = e^{\int \frac{2}{x} dx} = e^{2 \log x} = e^{\log x^2} = x^2$

Multiplying both sides by $x^2 \Rightarrow x^2 \frac{dy}{dx} + 2xy = x^2 \log x$

$\Rightarrow \frac{d}{dx}(yx^2) = x^2 \log x$; Integrating $\Rightarrow$

$x^2 y = \int x^2 \log x dx + C = C + \log x \cdot \frac{x^4}{4} - \int \frac{1}{x} \frac{x^4}{4} dx$

$\Rightarrow x^2 y = C + \frac{x^4}{4} \log x - \frac{1}{4} \cdot \frac{x^4}{4} = C + \frac{x^4}{4} \left(\log x - \frac{1}{4} \right)$

$\Rightarrow y = Cx^{-2} + \frac{1}{4}x^2 \left(\log x - \frac{1}{4} \right)$

which is the required solution

Solve the I.V.P $(x^2 + 1) \frac{dy}{dx} + 4xy = x$ O.C $y(2) = 1$ --- ②

Sol: ② $\Rightarrow \frac{dy}{dx} + \frac{4x}{x^2+1}y = \frac{x}{x^2+1}$ --- ③

$IF = e^{\int \frac{4x}{x^2+1} dx} = e^{2 \ln(x^2+1)} = e^{\ln(x^2+1)^2}$
 $= (x^2+1)^2$

Multiplying (3) by $(x^2+1)^2 \Rightarrow (x^2+1)^2 \frac{dy}{dx} + \frac{4x}{(x^2+1)} (x^2+1)^2 y$

$$\Rightarrow (x^2+1)^2 \frac{dy}{dx} + 4x(x^2+1)y = x(x^2+1)$$

$$\Rightarrow \frac{d}{dx} (x^2+1)^2 y = x^3 + x; \text{ Integrating,}$$

$$(x^2+1)^2 y = \int x^3 + x dx = \frac{x^4}{4} + \frac{x^2}{2} + C \dots (4)$$

Applying IC (2) $\Rightarrow (2^2+1)^2 \cdot 1 = \frac{2^4}{4} + \frac{2^2}{2} + C$

$\Rightarrow C = 19$ $\therefore (4) \Rightarrow (x^2+1)^2 y = \frac{x^4}{4} + \frac{x^2}{2} + 19$ Ans!

Bernoulli Equations: Eqn of the form $\frac{dy}{dx} + P(x)y = Q(x)y^n$ (1)

is called a Bernoulli DE.

From (1) $\Rightarrow y^{-n} \frac{dy}{dx} + P(x)y^{-n+1} = Q(x)$ (2)

Put $y^{-n+1} = v$, $\Rightarrow (-n+1)y^{-n} \frac{dy}{dx} = \frac{dv}{dx}$ so that,

$$(2) \Rightarrow \frac{1}{1-n} \frac{dv}{dx} + P(x)v = Q(x)$$

$$\Rightarrow \frac{dv}{dx} + (1-n)P(x)v = (1-n)Q(x)$$

which is a linear equation in v and x .

Eqn $f(y) \frac{dy}{dx} + P(x)f(y) = Q(x)$ (3)

Put $f(y) = v \Rightarrow f'(y) \frac{dy}{dx} = \frac{dv}{dx}$

$\therefore (1) \Rightarrow \frac{dv}{dx} + P(x)v = Q(x)$ which is linear in v and x .

Ques: Solve ① $\frac{dy}{dx} + xy = xy^2$, ② $\frac{dy}{dx} + y = xy^3$

Solⁿ: ① This is a Bernoulli's DE, when $n=3$, Dividing by

$$y^3 \text{ (Multiplying by } y^{-3}) \Rightarrow y^{-3} \frac{dy}{dx} + y^{-2} = x \quad \text{--- ①}$$

let, $+v = y^{1-n} = y^{1-3} = y^{-2}$ then, $\frac{dv}{dx} = -2y^{-3} \frac{dy}{dx}$

So that, $y^{-3} \frac{dy}{dx} = \frac{-1}{2} \frac{dv}{dx}$

The eqn ① $\Rightarrow -\frac{1}{2} \frac{dv}{dx} + v = x \Rightarrow \frac{dv}{dx} - 2v = -2x$ --- ②

IF = $e^{\int -2dx} = e^{-2x}$ Multiplying ② by $e^{-2x} \Rightarrow$

$$e^{-2x} \frac{dv}{dx} - 2ve^{-2x} = -2xe^{-2x} \Rightarrow \frac{d(v e^{-2x})}{dx} = -2xe^{-2x}$$

Integrating, $ve^{-2x} = \int -2xe^{-2x} dx + c$

$$= -2 \left[-\frac{1}{2} x e^{-2x} - \int \frac{1}{2} e^{-2x} dx \right] + c$$

$$= x e^{-2x} + \frac{1}{2} e^{-2x} + c = \frac{1}{2} (2x+1) e^{-2x} + c$$

$\Rightarrow v = \frac{1}{2} (2x+1) e^{2x}$ where c is an arbitrary const

But $v = \frac{1}{y^2}$ so that, we obtain the solⁿ in

the form, $\frac{1}{y^2} = x + \frac{1}{2} + \frac{e^{2x}}{2}$ Ans:

$y = \frac{1}{\sqrt{x + \frac{1}{2} + \frac{e^{2x}}{2}}}$

Solve $x \frac{dy}{dx} + y = y^2 \log x$

Solⁿ: Dividing by $xy^2 \Rightarrow y^{-2} \frac{dy}{dx} + \frac{1}{x} y^{-1} = \frac{1}{x} \log x$ (1)

Put, $-y^{-1} = v \Rightarrow \frac{dv}{dx} = -y^{-2} \frac{dy}{dx}$

$\therefore (1) \Rightarrow \frac{dv}{dx} - \frac{1}{x} v = -\frac{1}{x} \log x$ (2)

$\therefore I.F. = e^{\int -\frac{1}{x} dx} = \frac{1}{x}$

(2) $\Rightarrow \frac{1}{x} \frac{dv}{dx} - \frac{1}{x^2} v = -\frac{1}{x^2} \log x$

$$\Rightarrow \frac{d}{dx} \left(v \frac{1}{x} \right) = -\frac{1}{x^2} \log x$$

Integrating $\Rightarrow v \frac{1}{x} = C + \frac{1}{x^2} \log x$

Exact DE

Total Differential: If the fⁿ $F(x, y)$ has continuous first partial derivatives in a domain D , then the total differential DF is defined by,

$$dF(x, y) = \frac{\partial F(x, y)}{\partial x} dx + \frac{\partial F(x, y)}{\partial y} dy$$

$\forall (x, y) \in D$

Exact Differential: The expression $M(x, y) dx + N(x, y) dy$ is called an exact differential if

$\exists$ (there exists) $F(x, y)$ s.t. $dF(x, y) = M(x, y) dx + N(x, y) dy$

$dF(x, y) = M(x, y)dx + N(x, y)dy$ for all $(x, y) \in D$

That is, $M(x, y) = \frac{\partial F}{\partial x}$ & $N(x, y) = \frac{\partial F}{\partial y}$

Exact DE: If $M(x, y)dx + N(x, y)dy$ is an exact differential, then the differential eqn $M(x, y)dx + N(x, y)dy = 0$ is called an exact DE

Theorem (Necessary condition):

Consider the DE $M(x, y)dx + N(x, y)dy = 0$ where

$M(x, y)$ and $N(x, y)$ have continuous 1st partial derivatives

$\forall (x, y) \in D$.

* If the DE (1) is exact in D ,

then, $\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x} \quad \forall (x, y) \in D$

* Conversely, if $\frac{\partial M(x, y)}{\partial y} = \frac{\partial N(x, y)}{\partial x} \quad \forall (x, y) \in D$,

then the DE (1) is exact in D .

Working rule: If the DE $M(x, y)dx + N(x, y)dy = 0$

satisfies the condition $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ then it's exact

To integrate it,

(i) Integrate w.r. to x regarding y as constant

(ii) Find out those terms in N which are

free from x and integrate them w.r.t y .

(IV) Add the two expressions so obtained an equate the sum to an arbitrary constant.

This gives the general solⁿ of the given exact DE.

↳ Solve $(3x^2 + 4xy)dx + (2x^2 + 2y)dy = 0$

Solⁿ: Here, $M(x, y) = 3x^2 + 4xy$ & $N(x, y) = 2x^2 + 2y$

$$\therefore \frac{\partial M}{\partial y} = 4x \text{ and } \frac{\partial N}{\partial x} = 4x \therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}; \forall (x, y)$$

Thus the given eqn is exact in every rectangular domain D .

① Integrate M w.r.t x regarding y as constant

$$\int M(x, y) dx = \int (3x^2 + 4xy) dx = x^3 + 2x^2y$$

② The terms in $N(x, y)$ free from x as $2y$ and

integrating this w.r.t y we get, $\int 2y dy = y^2$

Adding these two & equating to a constant

$x^3 + 2x^2y + y^2 = c$ which is the required solⁿ.

Note: You may solve it as it's given in,

J. L. Ross.

Solve the IVP $(2x \cos y + 3x^2 y) dx + (x^3 - x^2 \sin y - y) dy = 0$
 $y(0) = 2$

Soln: $\frac{\partial M}{\partial y} = -2x \sin y + 3x^2 = \frac{\partial N}{\partial x}$

→ The given DE is exact, ① $\int (2x \cos y + 3x^2 y) dx = x^2 \cos y + x^3 y$

② $\int -y dy = -\frac{y^2}{2}$ Adding: $x^2 \cos y + x^3 y - \frac{y^2}{2} = C$

At $x=0, y=2$ $0 \cdot \cos 2 + 0 \cdot 2 + \frac{2^2}{2} = C \Rightarrow C = -2$

$\therefore x^2 \cos y + x^3 y - \frac{y^2}{2} = -2$ Ans

Integrating Factor: If the DE $M(x, y) dx + N(x, y) dy = 0$

is not exact but it has been multiplied by $M(x, y)$ that is

$M(x, y) M(x, y) dx + \mu(x, y) N(x, y) dy = 0$ — ②

$\mu(x, y)$ is called an integrating factor of ①

Rules for finding IF: R.1 If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$

is a f^n of x only, then $e^{\int f(x) dx}$ is an IF.

R.2 If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} = g(y)$, f^n of y only, then,

$e^{\int g(y) dy}$ is an IF.

R.3) If $M dx + N dy = 0$ is homogeneous and

$Mx + Ny \neq 0$ then $\frac{1}{Mx + Ny}$ is an IF.

Solve $(x^2 + y^2 + 1) dx - 2xy dy = 0$ — (1)

Solⁿ: $M = x^2 + y^2 + 1$, $N = -2xy$; $\frac{\partial M}{\partial y} = 2y$, $\frac{\partial N}{\partial x} = -2y$

$\therefore \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \Rightarrow$ DE (1) is not exact.

However, $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y - (-2y)}{-2xy} = \frac{-2}{x}$ a

fⁿ of x alone, $\therefore IF = e^{\int \frac{-2}{x} dx} = e^{-2 \log x} = \frac{1}{x^2}$

Multiplying by $\frac{1}{x^2}$, DE (1) $\Rightarrow (1 + \frac{y^2}{x^2} + \frac{1}{x^2}) dx - \frac{2y}{x^2} dy = 0$ — (2)

Now, $\frac{\partial M}{\partial y} = \frac{2y}{x^2}$ & $\frac{\partial N}{\partial x} = \frac{2y}{x^2} \Rightarrow$ DE is exact.

Now, Integrating $1 + \frac{y^2}{x^2} + \frac{1}{x^2}$ w.r.t. x, keeping as constant, we get, $x - \frac{y^2}{x} - \frac{1}{x}$ and, in $-\frac{2y}{x}$

there is no term free from y.

Hence the solⁿ is, $x - \frac{y^2}{x} - \frac{1}{x} = C$ Ans:

* Solve: $(x^2 + y^2 + 2x) dx + 2y dy = 0$ — (1)

Solⁿ: $\frac{\partial M}{\partial y} = 2y$ & $\frac{\partial N}{\partial x} = 0$ $\therefore IF = e^{\int 1 dx} = e^x$

Now, $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y - 0}{2y} = 1$

multiplying (1) by $e^x \Rightarrow e^x (x^2 + y^2 + 2x) dx + e^x 2y dy = 0$ which is exact

This eqn can be written as:

$$e^x (x^2 + 2x) dx + (y^2 e^x) dx + e^x 2y dy = 0$$

$$\Rightarrow d(x^2 e^x) + d(y^2 e^x) = 0 \Rightarrow e^x (x^2 + y^2) = c \quad \text{Ans.}$$

Integrating, we get, $x^2 e^x + y^2 e^x = c$

Trajectories

Let, $F(x, y, c) = 0 \dots (1)$ be a given one parameter family of curves.

A curve that intersects the curves of the family (i) at a constant angle $\alpha \neq 90^\circ$ is called an oblique trajectory (or α trajectory) of the given family.

A curve that intersects the curves of the family at right angles ($\alpha = 90^\circ$) is called an orthogonal trajectory of the given family.

Equations of the trajectory: If a family of curves be given by the DE $f(x, y, p) = 0$ then, it's a trajectory's given by,

$$f\left(x, y, \frac{p - \tan \alpha}{1 + p \tan \alpha}\right) = 0$$

Orthogonal trajectories: If $\alpha = 90^\circ$, then

$$\frac{p - \tan \alpha}{1 + p \tan \alpha} = \frac{p \cot \alpha - 1}{\cot \alpha + p} = \frac{-1}{p} \quad [\cot 90^\circ = 0]$$

Hence, corresponding to the family of curve whose DE is $f(x, y, p) = 0$, the differential eqn of the orthogonal trajectory is, $f(x, y, -p) = 0$ which on integration gives family of trajectories orthogonal to the given family of curves.

Polar coordinates: If a family of curves be given by $f(r, \theta, \frac{dr}{d\theta}) = 0$ then the orthogonal trajectory is given by $f(r, \theta, -r^2 \frac{d\theta}{dr}) = 0$

Examples: Find the orthogonal trajectories of the family of circles $x^2 + y^2 = c^2$ --- (1)

Soln: Family of circles is $x^2 + y^2 = c^2$ --- (1)

$$\text{Differentiating} \Rightarrow 2x + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{y} \text{ --- (2) which is the DE of the given family --- (1)}$$

We now replace $-\frac{x}{y}$ by its negative reciprocal $\frac{y}{x}$ in the DE (2) to obtain the DE (2).

to obtain the differential equation.

$\frac{dy}{dx} = \frac{y}{x}$ --- (3) the orthogonal trajectories.

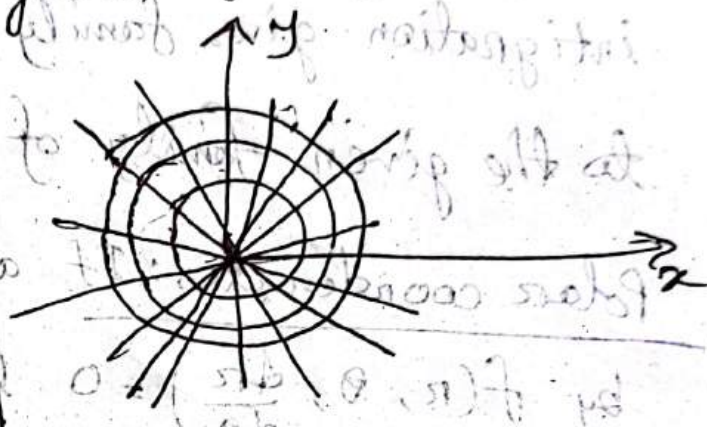
We now solve (3). Separating variables, we have

$$\frac{dy}{y} = \frac{dx}{x} \text{ Integrating} \Rightarrow \ln y = \ln x + \ln k$$

$$\therefore y = kx$$

This is one parameter family of solⁿs of D.E. (1) and thus represents the family of orthogonal trajectories of the given family of circles.

Find the orthogonal trajectories of $y = cx^2$



Solⁿ: Differentiating $y = cx^2$ w.r.t. x $\Rightarrow \frac{dy}{dx} = 2cx = 2 \cdot \frac{y}{x^2} \cdot x = \frac{2y}{x}$
 Replacing $\frac{dy}{dx}$ by $-\frac{x}{2y}$ we get $\frac{dy}{dx} = -\frac{x}{2y}$ which

is the D.E. of orthogonal trajectory $\Rightarrow 2y dy + x dx = 0$

Integrating $y^2 + \frac{x^2}{2} = k_0 \rightarrow \log k$ (separating the var^s)

$$\Rightarrow 2y^2 + x^2 = 2k_0 = k$$

which is the family of orthogonal trajectories

(2) of the given family $y = cx^2$.

Find a family of oblique trajectory of the family of straight lines $y = cx$ at angle $\alpha = 45^\circ$ (SL Rom P 75)

Higher order linear ODE: A. LODE of order n in the dependent variable y and the independent variable x is an equation that can be expressed in the form,

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \frac{dy}{dx} + a_n(x) y = F(x) \quad \text{--- (1)}$$

where $a_0(x) \neq 0$. We shall assume that $a_0, a_1, \dots, a_n$ are continuous on area interval $a \leq x \leq b$ & $a_0 \neq 0 \forall x$ on $a \leq x \leq b$. The right-hand member $F(x)$ is called the non-homogenous term.

If $F(x) = 0$ then eqn (1) reduces to

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \frac{dy}{dx} + a_n(x) y = 0 \quad \text{--- (2)}$$

and is then called homogeneous.

Let, x_0 be any point on $a \leq x \leq b$ and let, $c_0, c_1, \dots, c_{n-1}$ be n arbitrary constants.

There exists a unique solⁿ f^* of the eqn (1) s.t.

$$f(x_0) = c_0, f'(x_0) = c_1, \dots, f^{(n-1)}(x_0) = c_{n-1}$$

which is defined on $a \leq x \leq b$

Note: Let, f be a solⁿ of (2) then,

$$f(x_0) = 0, f'(x_0) = 0, \dots, f^{(n-1)}(x_0) = 0$$

Defⁿ: If $f_1, f_2, \dots, f_m$ are m f's and $c_1, c_2, \dots, c_m$ are m constants then the expression $c_1 f_1 + c_2 f_2 + \dots + c_m f_m$ is called a linear combination of $f_1, f_2, \dots, f_m$.

let, $f_1, f_2, \dots, f_m$ are m solⁿs of (2) then $c_1 f_1 + c_2 f_2 + \dots + c_m f_m$ is also solⁿ of (2).

Theorem: The n th order homogenous linear DE.

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \frac{dy}{dx} + a_n(x) y = 0$$

$= 0 \rightarrow$ always possess n solutions that are linear independent. Further if $f_1, f_2, \dots, f_n$ are

" n " linearly independent solⁿs of (2) then every solⁿ of (2) can be expressed as a linear combination: $c_1 f_1 + c_2 f_2 + c_3 f_3 + \dots + c_n f_n$

by proper choice of the constants $c_1, c_2, \dots, c_n$.

If $f_1, f_2, \dots, f_n$ are n linearly independent solⁿs of (2) then, $f(x) = c_1 f_1 + c_2 f_2 + \dots + c_n f_n, a \leq x \leq b$

& $c_1, c_2, \dots, c_n$ are constants, is called a general solⁿ of (2).

Wronskian:

Let, $f_1, f_2, \dots, f_n$ be n real f^n each of which has a $n(n-1)$ derivative on $a \leq x \leq b$, the determinant

$$W(f_1, f_2, \dots, f_n) = \begin{vmatrix} f_1 & f_2 & \dots & f_n \\ f_1' & f_2' & \dots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \dots & f_n^{(n-1)} \end{vmatrix}$$

in which primes denotes derivatives, is called the Wronskian of $f_1, f_2, \dots, f_n$ is different from

zero for some x on $a \leq x \leq b$.

Show that, the solⁿ, e^x, e^{-x} and e^{2x} of the

DE $\frac{d^3 y}{dx^3} - 2 \frac{d^2 y}{dx^2} - \frac{dy}{dx} + 2y = 0$ are linearly independent

Solⁿ: We find that the Wronskian:

$$W(e^x, e^{-x}, e^{2x}) = \begin{vmatrix} e^x & e^{-x} & e^{2x} \\ e^x & -e^{-x} & 2e^{2x} \\ e^x & e^{-x} & 4e^{2x} \end{vmatrix}$$

$$= e^x e^{-x} e^{2x} \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 1 & 1 & 4 \end{vmatrix} = e^{2x} (-6 + 2 - 2)$$

$$= -6e^{2x} \neq 0 \quad \forall x \in \mathbb{R}$$

Thus, since $W(e^x, e^{-x}, e^{2x}) \neq 0$ for all real values of x . We conclude that, e^x, e^{-x}, e^{2x} are independent

So, linearly independent solⁿ of the given DE on every real interval.

Reduction of Order:

Let, y be a non-trivial solⁿ of the n^{th} order

$$\text{HLDE (2)}: a_0(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + \dots + a_{n-1}(x) \frac{dy}{dx} + a_n(x)y = 0 \dots (2)$$

The transformation, $y = f(x)v$ reduces eqn (2) to an $(n-1)$ s.f. order homogenous linear DE in the dependent variable $w = \frac{dv}{dx}$

Note: For general discussion, (to find the solⁿ).

Please see SL. Ross P 116 - 117

Here we'll discuss this method by solving a problem.

as follows: Given that $y=x$ is a solⁿ of the DE:

$$(x^2+1) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$$

find a linear independent solⁿ by reducing the order. Write the general solⁿ.

Solⁿ: We are given $y=x$ is a solⁿ of

$$(x^2+1) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0 \dots (1)$$

Here $y=x$, $\frac{dy}{dx} = 1$, $\frac{d^2 y}{dx^2} = 0$

L.H.S. of (1) $\Rightarrow (x^2+1) \cdot 0 - 2x \cdot 1 + 2x = 0 \Rightarrow \text{R.H.S.}$

$\Rightarrow y=x$ satisfies the eqn (1)

let, $y = vx$ then, $\frac{dy}{dx} = x \frac{dv}{dx} + v$ & $\frac{d^2y}{dx^2} = x \frac{d^2v}{dx^2} + 2 \frac{dv}{dx}$

Substituting the expressions for y , $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ into

eqn(1) we obtain, $(x^2+1) \left(x \frac{d^2v}{dx^2} + 2 \frac{dv}{dx} \right) - 2x \left(x \frac{dv}{dx} + v \right)$

$$\Rightarrow x(x^2+1) \frac{d^2v}{dx^2} + 2(x^2+1) \frac{dv}{dx} - 2x^2 \frac{dv}{dx} - 2xv + 2xv = 0$$

$$\Rightarrow x(x^2+1) \frac{d^2v}{dx^2} + 2 \frac{dv}{dx} = 0 \dots \textcircled{II}$$

let, $w = \frac{dv}{dx}$ then, we get from $\textcircled{II}$, $x(x^2+1) \frac{dw}{dx} + 2w = 0$ which is a 1st order homogeneous linear eqn. $\textcircled{III}$

Treating this as a separable eqn, we obtain,

$$\frac{dw}{w} = \frac{-2dx}{x(x^2+1)} = \left(-\frac{2}{x} + \frac{2x}{x^2+1} \right) dx \quad [\text{By partial fraction}]$$

Integrating, we obtain, $\ln w = -2 \ln x + \ln(x^2+1) + c$

$$\Rightarrow w = c \frac{(x^2+1)}{x^2} \Rightarrow \frac{dv}{dx} = \frac{x^2+1}{x^2} \quad [\text{since, } w = \frac{dv}{dx} \text{ and we choose, } c=1]$$

Integrating, we obtain, $v = x - \frac{1}{x}$

Now, forming $y = xv$, where $f(x)$ denote the known solⁿ $x(y=f(x))$, we obtain the f^h g defined by,

$$g = xv = x \left(x - \frac{1}{x} \right) = x^2 - 1$$

which is the desired linearly independent solⁿ the general solⁿ of the given, eqn (1) is the linear

combination of f & g , and thus we write the general solⁿ of (1) as, $y = c_1 x + c_2 (x^2 - 1)$, c_1 & c_2 are arbitrary constants.

General and Particular Solⁿ:

Consider the n th order (non homogenous) linear DE

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \frac{dy}{dx} + a_n(x) y = Q(x) \quad (2)$$

1) The general solⁿ of (2) is called the complementary fⁿ of eqn (1) (CF) and we shall denote this by y_c .

2) Any particular solⁿ of eqn (1) involving no arbitrary constants is called a particular integral (PI) of (1) and we denote this by y_p .

3) The solⁿ $y_c + y_p$ is called the general solⁿ of (1) that is the general solⁿ $= y_c + y_p = CF + PI$.

The operator D . It's usual to write, $D = \frac{d}{dx}$.

$$D^2 = \frac{d^2}{dx^2}, \quad D^n = \frac{d^n}{dx^n}$$

In terms of the operator D the DE (1) can be written as $(a_0 D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_{n-1} D + a_n) y = F(x)$

Note: It can be proved that, D can be treated as algebraic quantity in several respects.

The homogenous linear equations with constant coefficients:

Consider the homogenous linear DE:

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = 0 \quad \text{--- (3)}$$

where $a_0, a_1, \dots, a_{n-1}, a_n$ are real constants. The general solⁿ of eqn (3) can be found explicitly.

[Note: we have to choose a f^n having the property that $\frac{d^k}{dx^k} f(x) = c f(x) \forall x$ and this is possible if, $f(x) = e^{mx}$]

let, $y = e^{mx}$ be a solⁿ of (3). Then putting, $y = e^{mx}$

$$\frac{dy}{dx} = m e^{mx} = D e^{mx}, \quad \frac{d^2 y}{dx^2} = m^2 e^{mx} = D^2 e^{mx}, \quad \dots, \quad \frac{d^n y}{dx^n} = D^n y = m^n e^{mx}$$

$m^n e^{mx}$ in (3) we get,

$$a_0 m^n e^{mx} + a_1 m^{n-1} e^{mx} + \dots + a_{n-1} m e^{mx} + a_n e^{mx} = 0$$

$$\text{or, } e^{mx} (a_0 m^n + a_1 m^{n-1} + \dots + a_{n-1} m + a_n) = 0$$

Since, $e^{mx} \neq 0$, we obtain the polynomial eqn.

in the unknown m :

$$a_0 m^n + a_1 m^{n-1} + \dots + a_{n-1} m + a_n = 0 \quad \dots \text{--- (4)}$$

This eqn (4) is called the auxiliary eqn or the characteristic eqn of the DE (3).

Hence, $y = e^{mx}$ will be solⁿ of (3) if m is a root of the algebraic eqn (4).

Note: The auxiliary eqn $f(m) = 0$ gives the same value of m as the eqn $f(D) = 0$ gives of D . Here,

$$f(D) = 0, \text{ i.e. } a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n = 0$$

can in general be regarded as the auxiliary eqn

Solution of LAIDE with constant coefficients:

Case 1: Distinct Real Roots of AE:

If $m_1, m_2, \dots, m_n$ be the n different real roots of the AE (4) then the general solⁿ of (3) is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}$$

Ex: $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = 0$ — (1)

Let, $y = e^{mx}$ be a solⁿ of (1) the AE is,

$$m^2 - 3m + 2 = 0 \Rightarrow (m-2)(m-1) = 0$$

$$\therefore m = 2, 1; m_1 = 2, m_2 = 1$$

The roots are real and distinct. Thus e^{2x} and

e^{2x} are solⁿs and the general solⁿ may be written as,
 $y = c_1 e^x + c_2 e^{2x}$

Verification: $y_1 = e^x$ & $y_2 = e^{2x}$ are 2 solⁿs, since,

$$W(e^x, e^{2x}) = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = e^{3x} \neq 0$$

$\Rightarrow y_1 = e^x$ & $y_2 = e^{2x}$ are 2 linearly independent solⁿs of the DE (I).

Ex: $\frac{d^3 y}{dx^3} - 4 \frac{d^2 y}{dx^2} + \frac{dy}{dx} + 6y = 0$ — (I)

AE: $m^3 - 4m^2 + m + 6 = 0$ — (II)

we observe that, $m = -1$ is a solⁿ of (II) so, that, $m+1=0$

is factor of (II), hence, $m^2(m+1) - 5m(m+1) + 6(m+1) = 0$

$$\Rightarrow (m+1)(m^2 - 5m + 6) = 0 \Rightarrow (m+1)(m-2)(m-3) = 0$$

$\Rightarrow m = -1, 2, 3$ which are real and distinct.

$\therefore m = -1, m = 2, m = 3$. Thus the general solⁿ is,

$$y = c_1 e^{-x} + c_2 e^{2x} + c_3 e^{3x} \quad \text{Ans.}$$

Case 2: Repeated Real Root:

(I) If the AE has two equal roots then the g.s.
 is $y = (c_1 + x c_2) e^{m_1 x}$ [$m_1 = m_2 = m$]

(II) Two repeated & others are distinct then,
 $y = (c_1 + c_2 x) e^{m_1 x} + c_3 e^{m_2 x} + c_4 e^{m_3 x} + \dots$

(III) The repeated roots then,
 $y = (c_1 + c_2 x + c_3 x^2) e^{m_1 x}$ [$m_1 = m_2 = m_3 = m$]

IV) m repeated roots occurring k times, then,
 $y = (c_1 + c_2 x + c_3 x^2 + \dots + c_k x^{k-1}) e^{mx}$

Ex: $\frac{d^3 y}{dx^3} - 4 \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 18y = 0$

AE: $m^3 - 4m^2 - 3m + 18 = 0 \Rightarrow (m+2)(m-3)(m-3) = 0$

$\therefore m = -2, 3$ the g.s. in, $y = (c_1 + c_2 x) e^{3x} + c_3 e^{-2x}$

Ex: $\frac{d^4 y}{dx^4} - 5 \frac{d^3 y}{dx^3} + 6 \frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} - 8y = 0$

The AE: $m^4 - 5m^3 + 6m^2 + 4m - 8 = 0$

The roots are $2, 2, 2, -1$ g.s. in, $y = (c_1 + c_2 x + c_3 x^2) e^{2x} + c_4 e^{-x}$

Case 3: Complex conjugate Roots (AE having Imaginary roots)

If the AE has imaginary roots in the form $a \pm ib$

then the g.s. in, $y = e^{ax} (c_1 \cos bx + c_2 \sin bx)$

If the Imaginary roots are repeated then the

g.s. in $y = e^{ax} \left[(c_1 + c_2 x + c_3 x^2 + \dots + c_k x^{k-1}) \sin bx + (c_{k+1} + c_{k+2} x + \dots + c_{2k} x^{k-1}) \cos bx \right]$

Ex: $\frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} + 25y = 0$

AE: $m^2 - 6m + 25 = 0 \Rightarrow m = \frac{6 \pm \sqrt{36 - 100}}{2}$

$\therefore m = \frac{6 \pm 8i}{2} = 3 \pm 4i$

The g.s. of $y = e^{3x} (c_1 \sin 4x + c_2 \cos 4x)$

$$\text{Ex: } \frac{d^4 y}{dx^4} - 4 \frac{d^3 y}{dx^3} + 14 \frac{d^2 y}{dx^2} - 20 \frac{dy}{dx} + 25y = 0$$

$$\text{AE: } m^4 - 4m^3 + 14m^2 - 20m + 25 = 0, \text{ Roots: } m = 1 \pm 2i$$

$$\therefore \text{gs } y = e^x \{ (c_1 + c_2 x) \sin 2x + (c_3 + c_4 x) \cos 2x \}$$

$$\# \text{ Solve the IVP, } \frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} + 25y = 0 \quad \text{--- (I)}$$

$$y(0) = -3 \quad \text{--- (I-a)}, \quad y'(0) = -1 \quad \text{--- (I-b)}$$

Let, $y = e^{mx}$ be a solⁿ of (I) then the

$$\text{Solⁿ: AE of (I) is } m^2 - 6m + 25 = 0 \Rightarrow m = \frac{6 \pm \sqrt{36 - 100}}{2}$$

$$\therefore m = \frac{6 \pm 8i}{2} = 3 \pm 4i \quad \text{--- The gs of (I) is,}$$

$$y = e^{3x} (c_1 \sin 4x + c_2 \cos 4x) \quad \text{--- (II)}$$

Applying IC (I-a): $y(0) = -3$ in (II) we get,

$$-3 = e^{3 \cdot 0} (c_1 \sin 4 \cdot 0 + c_2 \cos 4 \cdot 0) \Rightarrow c_2 = -3$$

Again, applying IC (I-b): $y'(0) = -1$ in (II) $\Rightarrow$

$$-1 = 3e^{3 \cdot 0} (c_1 \sin 4 \cdot 0 + c_2 \cos 4 \cdot 0) + e^{3 \cdot 0} (4c_1 \cos 4 \cdot 0 - 4c_2 \sin 4 \cdot 0)$$

$$\Rightarrow -1 = 3c_2 + 4c_1 \Rightarrow -1 = 3(-3) + 4c_1, \quad \because c_2 = -3$$

$$\Rightarrow c_1 = 2 \quad \text{Replacing the value of } c_1 \text{ \& } c_2 \text{ in}$$

$$\text{(II) we get, } y = e^{3x} \{ (2 \sin 4x) - (3 \cos 4x) \}$$

which is the unique solⁿ of the given IVP.

Meaning of the symbol $\frac{1}{f(D)}$

Defⁿ: $\frac{1}{f(D)} F(x)$ is that f^n of x , free from arbitrary constant, which when operated by $f(D)$ gives,

$$F(x). \text{ Thus, } f(D) \frac{1}{f(D)} F(x) = F(x)$$

$\Rightarrow f(D)$ and $\frac{1}{f(D)}$ are inverse operator. Thus, the symbol $\frac{1}{D}$ stands for integration

Particular Integral in some special case:

Particular integral solⁿ when, $F(x) = e^{ax}$

$$\# \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax} \quad (\text{provided, } f(a) \neq 0)$$

Ex: Solve, $\frac{d^2y}{dx^2} + 2k \frac{dy}{dx} + k^2 y = e^x$

Solⁿ: A.E: $m^2 + 2km + k^2 = 0 \Rightarrow (m+k)^2 = 0$

$\therefore m = -k, -k \quad \therefore C.F. = (C_1 + C_2 x) e^{-kx}$

P.I. = $\frac{1}{f(D)} e^x = \frac{1}{D^2 + 2kD + k^2} e^x \quad [m = 1]$

$$= \frac{1}{1^2 + 2k(1) + k^2} e^x = \frac{e^x}{(1+k)^2}$$

Hence the Integral solⁿ is, $y = C.F. + P.I.$

$$\Rightarrow y = (C_1 + C_2 x) e^{-kx} + \frac{e^x}{(1+k)^2}$$

$$\# \frac{1}{f(D^2)} \sin(ax) = \frac{1}{f(-a^2)} \sin(ax) \text{ if } f(-a^2) \neq 0$$

$$\# \frac{1}{f(D^2)} \cos(ax) = \frac{1}{f(-a^2)} \cos(ax), \text{ if } f(-a^2) \neq 0$$

Ex: ~~AE~~ $\frac{d^3 y}{dx^3} + \frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = \sin 2x$

Soln: ~~AE~~ $\Rightarrow m^3 + m^2 + m + 1 = 0 \Rightarrow (m^2 + 1)(m + 1) = 0$

$m = -1, \pm i$

CF = $c_1 e^x + c_2 \cos x + c_3 \sin x$

PI = $\frac{1}{f(D)} \sin 2x = \frac{1}{(D^2 + 1)(D + 1)} \sin 2x$

$= \frac{1}{(2^2 + 1)(D + 1)} \sin 2x = \frac{1}{3} \frac{(D - 1) \sin 2x}{(D + 1)(D - 1)}$

$= \frac{1}{3} \frac{D - 1}{D^2 - 1} \sin 2x = \frac{1}{3} \frac{D - 1}{(-2^2 - 1)} \sin 2x$

$= \frac{1}{15} (2 \cos 2x - \sin 2x)$

$\therefore$ The general complete soln is, CF + PI.

$\Rightarrow y = c_1 e^x + c_2 \cos x + c_3 \sin x + \frac{1}{15} (2 \cos 2x - \sin 2x)$

Exceptional cases: $\frac{1}{f(D)} e^{ax}$ when $f(a) = 0$

$\frac{1}{f(D)} e^{ax} = \frac{x e^{ax}}{f'(a)}$ if $f'(a) \neq 0$

$\frac{1}{f(D)} e^{ax} = \frac{x^2 e^{ax}}{f''(a)}$ if $f'(a) = 0$ but $f''(a) \neq 0$ and so on

$$\star \text{ If } f(-a) = 0 \Rightarrow \frac{1}{f(D^2)} \sin ax = \frac{x}{f'(D^2)} \sin(ax)$$

$$\Rightarrow \frac{1}{f(D^2)} \cos ax = \frac{x}{f'(D^2)} \sin(ax)$$

$$\text{Ex: } \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^x$$

$$\text{AE: } m^2 - 3m + 2 = 0 \Rightarrow (m-2)(m-1) = 0 \therefore m = 1, 2$$

$$\therefore \text{CF} = c_1 e^x + c_2 e^{2x} \quad \therefore \text{PI} = \frac{e^x}{(D-2)(D-1)} = \frac{x \cdot e^x}{2D-3}$$

(case of failure $\leftarrow f(a) = 0$) $\therefore \text{PI} = \frac{x e^x}{e \cdot 1 - 3} = -x e^x$

$$\therefore \text{gs: } y = \text{CF} + \text{PI} = c_1 e^x + c_2 e^{2x} - x e^x$$

Evaluating $\frac{1}{f(D)} x^m$ expand $[f(D)]^{-1}$ in ascending powers of D .

$$\text{Ex: } \frac{d^3 y}{dx^3} + 2 \frac{d^2 y}{dx^2} + \frac{dy}{dx} = e^{2x} + x^2 + x$$

$$\text{AE: } m^3 + 2m^2 + m = 0 \Rightarrow m(m+1)^2 = 0 \therefore m = 0, -1, -1$$

$$\therefore \text{CF} = c_1 (c_2 + c_3 x) e^{-x}$$

$$\text{PI} = \frac{e^{2x}}{D(D+1)^2} + \frac{1}{D(D+1)^2} (x^2 + x)$$

$$= \frac{e^{2x}}{2(2+1)^2} + \frac{1}{D} (1+D)^{-2} (x^2 + x)$$

$$= \frac{e^{2x}}{18} + \frac{1}{D} (1 - 2D + 3D^2 - \dots) (x^2 + x)$$

$$= \frac{e^{2x}}{18} + D^{-1} (x^2 + x - 2(2x+1) + 3(2) + 0)$$

$$= \frac{e^{2x}}{18} + \frac{1}{6}(x^2 + x - 4x - 2 + 6) = \frac{e^{2x}}{18} + \frac{1}{6}(x^2 - 3x + 4)$$

$$= \frac{e^{2x}}{18} + \frac{x^3}{3} - \frac{3x^2}{2} + 4x$$

$$\therefore \text{gen. sol. } y = CF + PI = (c_1 + (c_2 + c_3 x)e^{-x} + \frac{e^{2x}}{18} + \frac{x^3}{3} - \frac{3x^2}{2} + 4x)$$

$$\# \frac{1}{f(D)}(e^{ax} v) = e^{ax} \frac{1}{f(D+a)} v$$

$$\# \frac{1}{f(D)}(x^m v) = \frac{1}{f(D)}(x^m v)$$

End of sheet

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From sheet 2

The method of undetermined coefficients:

Defⁿ: consider an Undetermined Coefficient (UC) function f . The set of functions consisting of f itself and linearly independent UC functions of which the successive derivatives of f are either constant multipliers or linear combinations will be called the UC set of f .

Ex! Solve by using MUC.

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 2e^x - 10 \sin x$$

Solⁿ: we are given, $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 2e^x - 10 \sin x$ — (1)

The corresponding homogeneous equation is.

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 0 \quad \text{--- (2)}$$

AE: $m^2 - 2m - 3 = 0$

$$\Rightarrow (m-3)(m+1) = 0 \Rightarrow m = 3, -1$$

The complementary function is, $y_c = c_1 e^{3x} + c_2 e^{-x}$ ③

The nonhomogeneous term is the linear combination

$2e^x - 10 \sin x$ of the two UC functions given by,

e^x and $\sin x$

1. Form the UC set for each of these two f's

we find, $S_1 = \{e^x\}$, $S_2 = \{\sin x, \cos x\}$

2. Note that neither of these sets are identical with nor included in the other, hence both are retained.

3. Furthermore, by examining the complementary f's, we see that none of the f's e^x , $\sin x$, $\cos x$ in either of these sets is a sol'n of the corresponding homogeneous eqn. Hence neither set needs to be revised.

4. Thus the original sets S_1 and S_2 remain intact in this problem, and we form the linear combination $Ae^x + B \sin x + C \cos x$

of the functions e^x , $\sin x$, $\cos x$ of S_1, S_2 with the undetermined coefficients A, B, C .

5. We determine these unknown coefficients by taking $y_p = Ae^x + B\sin x + C\cos x$ — (4)

As a particular solⁿ of (1) then,

$$y_p' = Ae^x + B\cos x - C\sin x$$

$$y_p'' = \cancel{Ae^x} + \cancel{B\sin x} + \cancel{C\cos x} = Ae^x - B\sin x - C\cos x$$

Substituting in (1) we get,

$$(Ae^x - B\sin x - C\cos x) - 2(Ae^x + B\cos x - C\sin x) - 3(Ae^x + B\sin x + C\cos x) = 2e^x - 10\sin x.$$

Since the eqn is to satisfy the DE (1) hence the coefficients of like terms of both sides must be respectively equal. Equating the coefficients of like terms, we obtain

$$\text{the eqns, } -4A = 2, \quad -4B + 2C = -10, \quad A + 2B = 0$$

Solving these eqn's, we get, $A = -\frac{1}{2}, B = 2, C = 1$

and hence we obtain the particular integral

$$y_p = -\frac{1}{2}e^x + 2\sin x - \cos x.$$

Thus the general solⁿ of the given DE is,

$$y = y_c + y_p = C_1 e^{2x} + C_2 e^{-x} - \frac{1}{2} e^x + 2 \sin x - \cos x$$

Ex 2: Solve: $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = 2x^2 + e^x + 2xe^x + 4e^{3x}$ — (1)

Solⁿ: The corresponding H.E. is, $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = 0$

The A.E. is $m^2 - 3m + 2 = 0 \Rightarrow (m-2)(m-1) = 0$
 $\therefore m = 2, 1$

and the C.F. is, $y_c = C_1 e^x + C_2 e^{2x}$

The non homogenous term in the linear combination

$2x^2 + e^x + 2xe^x + 4e^{3x}$ of the U.C fⁿs given

by x^2, e^x, xe^x, e^{3x}

1. Form the U.C set for each of these fⁿs, we have

$S_1 = \{x^2, x, 1\}; S_2 = \{e^x\}; S_3 = \{xe^x, e^x\},$

$S_4 = \{e^{3x}\}$ ~~$3e^x$~~

2. We note that, S_2 is completely included in S_3
 so S_2 is omitted from further consideration,

leaving the three sets, $S_1 = \{x^2, x, 1\},$

$S_3 = \{xe^x, e^x\}, S_4 = \{e^{3x}\}$

3. We now observe that $S_3 = \{xe^x, e^x\}$ includes e^x (which is included in the CF and so is a solⁿ of the corresponding HDE. Thus we multiply each member of S_3 by x to obtain the revised family, $S_3' = \{x^2e^x, xe^x\}$ which contains no members that are solⁿ of the corresponding HDE (2).

4. Thus we have UC sets, $S_1 = \{x^2, x, 1\}$,

$$S_2 = \{e^{3x}\}, S_3' = \{x^2e^{3x}, xe^{3x}\}$$

We form the linear combination,

$$Ax^2 + Bx + C + De^{3x} + Ex^2e^x + Fxe^x$$

of the basis elements with UC A, B, C, D, E, F

5) We determine these unknown by taking particular solⁿ, $y_p = Ax^2 + Bx + C + De^{3x} + Ex^2e^x + Fxe^x$

$$y_p' = 2Ax + B + 3De^{3x} + Ex^2e^x + Fxe^x$$

From this we have,

$$y_p'' = 2A + 9De^{3x} + 2Exe^x + Fe^x$$

$$y_p'' = 2A + 9De^{3x} + 2Exe^x + Fe^x$$

$$y_p'' = 2A + 9De^{3x} + 2Exe^x + Fe^x$$

+ $Fxe^x + 2Fe^x$. Substituting y_p, y_p', y_p'' in the DE (1)

for y, y', y'' respectively, we obtain,

$$2A + 9D e^{3x} + E x^2 e^x + (4E + F) x e^x + (2E + 2F) e^x - 3[2Ax + B + 3D e^{3x} + E x^2 e^x + (2E + F) x e^x + F e^x] + 2[Ax^2 + Bx + C + D e^{3x} + E x^2 e^x + F x e^x] =$$

$$2x^2 + e^x + 2x e^x + 4e^{3x}$$

$$\text{or, } (2A - 3B + 2C) + (2B - 6A)x + 2Ax^2 + 2D e^{3x} + (-2E)x e^x + (2E - F)e^x = 2x^2 + e^x + 2x e^x + 4e^{3x}$$

Equating the coefficients of like terms -

$$2A - 3B + 2C = 0$$

$$2B - 6A = 0, \quad 2D = 4$$

$$2A = 2, \quad -2E = 2$$

$$2E - F = 1$$

Solving these eqns we get, $A = 1, B = 3, C = \frac{7}{2}$

$$D = 2, E = -1, F = -3$$

So the particular integral is,

$$y_p = x^2 + 3x + \frac{7}{2} + 2e^{3x} - x^2 e^x - 3x e^x$$

The general solⁿ is, therefore,

$$y = y_c + y_p = c_1 e^x + c_2 e^{-x} + x^2 + 3x + \frac{7}{2} + 2e^{3x} - x^2 e^x - 3x e^x$$

Solve $\frac{d^4 y}{dx^4} + \frac{d^2 y}{dx^2} = 3x^2 + 4 \sin x - 2 \cos x$

Solve the initial value problem, $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 2e^x - 10 \sin x$

Soln: $= 2e^x - 10 \sin x$

We are given $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 2e^x - 10 \sin x$

The corresponding homogeneous function, $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 0$

Auxiliary Equation: $m^2 - 2m - 3 = 0$

$\Rightarrow (m-3)(m+1) = 0 \therefore m = 3, -1$

$\therefore$ The complementary function, $y_c = c_1 e^{3x} + c_2 e^{-x}$

The non homogeneous fⁿ $2e^x - 10 \sin x$ is term in the

linear combination $2e^x - 10 \sin x$ of the VC fⁿ

given by $e^x, \sin x$.

1) form the VC sets for each these fⁿ, we have,

$S_1 = \{e^x\}, S_2 = \{\sin x, \cos x\}$

2) The

Solve the initial value problem $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 2e^x - 10 \sin x$
 $y(0) = 2, y'(0) = 4$

Hints: We have, $y = c_1 e^{3x} + c_2 e^{-x} - \frac{1}{2} e^x + 2 \sin x - \cos x$

$\therefore \frac{dy}{dx} = 3c_1 e^{3x} - c_2 e^{-x} - \frac{1}{2} e^x + 2 \cos x + \sin x$

Applying IC $\Rightarrow 2 = c_1 + c_2 - \frac{1}{2} \Rightarrow c_1 + c_2 = \frac{5}{2}$

$4 = 3c_1 - c_2 - \frac{1}{2} \Rightarrow 3c_1 - c_2 = \frac{9}{2}$

Solving $\Rightarrow c_1 = \frac{3}{2}, c_2 = 2$

Hence, $y = \frac{3}{2} e^{3x} + 2e^{-x} - \frac{1}{2} e^{2x} + 2 \sin x - \cos x$ Ans

Variation of Parameters:

Ex: Solve, $\frac{d^2 y}{dx^2} + y = \tan x$ --- (1)

Soln: The CF of (1) is $y_c = e_1 \sin x + e_2 \cos x$

We assume, $y_p = v_1(x) \sin x + v_2(x) \cos x$ --- (2)

where the functions v_1 & v_2 will be determined such that this is a particular integral of the

DE (1). Then, $y_p' = v_1(x) \cos x - v_2(x) \sin x$

We impose the condition $v_1'(x) \sin x + v_2'(x) \cos x = 0$ --- (3)

So that, $y_p' = v_1(x) \cos x - v_2(x) \sin x$

$\Rightarrow y_p'' = -v_1(x) \sin x - v_2(x) \cos x + v_1'(x) \cos x - v_2'(x) \sin x$ --- (4)

Substituting (2) and (4) into (1) we obtain;

$v_1'(x) \cos x - v_2'(x) \sin x = \tan x$ --- (5)

Thus we have two equations (3) and (5) from which

to determine $v_1'(x)$, $v_2'(x)$

$v_1'(x) \sin x + v_2'(x) \cos x = 0$ --- (3) $v_1'(x) \cos x - v_2'(x) \sin x = \tan x$ --- (5)

Solving we find; $v_1'(x) = \frac{\begin{vmatrix} 0 & \cos x \\ \tan x & -\sin x \end{vmatrix}}{\begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix}} = \frac{-\cos x \tan x}{-\sin^2 x - \cos^2 x} = \sin x$

Continue after
1st sheet

$$v_2'(x) = \frac{\begin{vmatrix} \sin x & 0 \\ \cos x & \tan x \end{vmatrix}}{\begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix}} = \frac{\sin x \tan x}{-\sin^2 x - \cos^2 x} = \frac{-\sin^2 x}{\cos x}$$

$$= \frac{\cos^2 x - 1}{\cos x} = \cos x - \sec x$$

Integrating, we find, $v_1(x) = \int \sin x dx = -\cos x + c_3$

$$v_2 = \int (\cos x - \sec x) dx = \sin x - \ln|\sec x + \tan x| + c_4$$

Substituting (6) into (2) we get,

$$y_p = (-\cos x + c_3) \sin x + (\sin x - \ln|\sec x + \tan x| + c_4) \cos x$$

$$= \sin x \cos x + c_3 \sin x + \sin x \cos x - \ln|\sec x + \tan x| (\cos x) + c_4 \cos x$$

$$= c_3 \sin x + c_4 \cos x - \cos x (\ln|\sec x + \tan x|)$$

Since the particular integral is a solⁿ free of arbitrary constants, we may assign any particular values A and B to c_3 and c_4 , respectively and thus, the result will be the PI:

$$y_p = A \sin x + B \cos x - \cos x (\ln|\sec x + \tan x|)$$

$$\text{Then, } y = y_c + y_p = c_1 \sin x + c_2 \cos x + A \sin x + B \cos x - \cos x (\ln|\sec x + \tan x|)$$

$$= (c_1 + A) \sin x + (c_2 + B) \cos x - \cos x (\ln|\sec x + \tan x|)$$

$\Rightarrow y = c_1 \sin x + c_2 \cos x - \cos x (\ln|\sec x + \tan x|)$ which is the general solⁿ of (1) where,
 $c_1 = c_1 + A$ and, $c_2 = c_2 + B$

Solve, $\frac{d^3 y}{dx^3} - 6 \frac{d^2 y}{dx^2} + 11 \frac{dy}{dx} - 6y = e^x$

by the method of variation of parameters.

$(x^2+1) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = 6(x^2+1)^2$

The Cauchy Cauchy - Euler Eqn / Homogenous linear eqn:

An eqn of the form, $a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_2 x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = F(x) \dots \textcircled{1}$

where, $a_0, a_1, a_2, \dots, a_n$ are constants & F is a fⁿ of x , is called the Cauchy - Euler eqn or equi-dimensional eqn or homogenous linear eqn. In this eqn each term in the left member is, a constant multiple of an expression of the form

$$x^k \frac{d^k y}{dx^k}$$

Important substitution / transformation: The transformation

$x = e^t$ reduces the $\textcircled{1}$ to linear DE with constant coefficients.

Consider 2nd order Cauchy - Euler eqn:

$$a_0 x^2 \frac{d^2 y}{dx^2} + a_1 x \frac{dy}{dx} + a_2 y = F(x) \text{ --- } \textcircled{2}$$

let, $x = e^t$, $x > 0$ we have. $t = \ln x$

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{1}{x} \frac{dy}{dt} \Rightarrow \frac{dy}{dt} = x \frac{dy}{dx}$$

$$\text{and, } \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dt} \right)$$

$$= \frac{1}{x} \frac{d}{dx} \left(\frac{dy}{dt} \right) + \frac{dy}{dt} \frac{d}{dx} \left(\frac{1}{x} \right)$$

$$= \frac{1}{x} \frac{d}{dt} \left(\frac{dy}{dt} \right) \frac{dt}{dx} + \frac{dy}{dt} \left(-\frac{1}{x^2} \right) = \frac{1}{x} \frac{d^2y}{dt^2} \frac{dt}{dx} - \frac{1}{x^2} \frac{dy}{dt}$$

$$= \frac{1}{x} \left(\frac{d^2y}{dt^2} \right) \cdot \frac{1}{x} - \frac{1}{x^2} \frac{dy}{dt}$$

$$\Rightarrow \frac{d^2y}{dx^2} \cdot x^2 = \frac{d^2y}{dt^2} - \frac{dy}{dt}$$

Substituting in eqn (2) $\Rightarrow$

$$a_0 \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) + a_1 \frac{dy}{dt} + a_2 y = F(e^t)$$

$$\Rightarrow A_0 \frac{d^2y}{dt^2} + A_1 \frac{dy}{dt} + A_2 y = G(t)$$

$$\text{where } A_0 = a_0, A_1 = a_1 - a_0, A_2 = a_2 \quad \dots G(t)$$

The proof in the general n^{th} order
can proceed in a similar fashion.

$$\text{Solve: } x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = x^3$$

See SL Rom

Ex: $x^3 \frac{d^3 y}{dx^3} - 4x^2 \frac{d^2 y}{dx^2} + 8x \frac{dy}{dx} - 8y = 4 \ln x \dots (1)$

Soln: Assuming, $x > 0$, we let, $x = e^t$, then,

$$t = \ln x \text{ and } \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \frac{dy}{dt}$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dt} \right)$$

$$= \frac{1}{x} \frac{d^2 y}{dt^2} \frac{dt}{dx} + \frac{dy}{dt} \cdot \frac{d}{dx} \left(\frac{1}{x} \right)$$

$$= \frac{1}{x^2} \frac{d^2 y}{dt^2} - \frac{1}{x^2} \frac{dy}{dt} = \frac{1}{x^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right)$$

$$\frac{d^3 y}{dx^3} = \frac{1}{x^2} \frac{d}{dx} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) + \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \frac{d}{dx} \left(\frac{1}{x} \right)$$

$$= \frac{1}{x^2} \left(\frac{d^3 y}{dt^3} \frac{dt}{dx} - \frac{d^2 y}{dt^2} \frac{dt}{dx} \right) - \frac{2}{x^3} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right)$$

$$= \frac{1}{x^3} \left(\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} \right)$$

Thus substituting in (1) $\Rightarrow$

$$\left(\frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} \right) - 4 \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) + 8 \frac{dy}{dt} - 8y = 4t$$

$$\Rightarrow \frac{d^3 y}{dt^3} - 7 \frac{d^2 y}{dt^2} + 14 \frac{dy}{dt} - 8y = 4t \dots (2)$$

The CF of (2) is, $y_c = c_1 e^t + c_2 e^{2t} + c_3 e^{4t}$

To find the PI of (2) we use method of undetermined coefficients

$$\begin{aligned} m^3 - 7m^2 + 14m - 8 &= 0 \\ (m-1)(m-2)(m-4) &= 0 \\ m &= 1, 2, 4 \end{aligned}$$

VC set, $s = \{t, 1\}$

We assume, $y_p = At + B$ then, $y_p' = A$, $y_p'' = 0$

$$\Rightarrow 14A - 8(At + B) = 4t$$

$$\Rightarrow -8A = 4 \text{ \& } 14A - 8B = 0 \Rightarrow A = -\frac{1}{2}, B = \frac{7}{8}$$

Thus the PI of (1) is $y_p = -\frac{1}{2}t + \frac{7}{8}$

and the GS of (2) is,

$$y = y_c + y_p = c_1 e^t + c_2 e^{2t} + c_3 e^{4t} - \frac{1}{2}t + \frac{7}{8}$$

and hence the GS of (1) is

$$y = c_1 x + c_2 x^2 + c_3 x^4 + \frac{1}{2} \ln x = -\frac{7}{8} \text{ Ans.}$$

$$\left(\frac{1}{x} \right) \frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}$$

① is particular solution

$$y_p = -\frac{1}{2}t + \frac{7}{8}$$

②
 $y'' + 2y' + 2y = 0$
 $(s^2 + 2s + 2)Y = 0$
 $(s+1)^2 + 1 = 0$
 $(s+1) = \pm i$
 $s = -1 \pm i$
 $y = e^{-x} (C_1 \cos x + C_2 \sin x)$